Model Collapse on Manifolds: a Closed-Form Capacity Floor and a Data-Anchoring Fix

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Abstract

Generative models are increasingly trained on data that includes their own past outputs, and for diffusion models this *self-consuming* loop is known to degrade quality from one generation to the next. The sharpest existing theory [1] shows that a carefully annealed early-stopping schedule repairs the loop, but only when the data density is smooth, explicitly leaving open the case that matters most in practice, where data concentrate on a thin low-dimensional manifold. We settle this case, in two parts. *Negatively*, we prove a *capacity floor*: near a manifold of width σ the score a diffusion model must represent is steep only inside a short, finite window of noise levels, so any network whose slope is capped below that peak bakes in a fixed amount of excess width on every pass that later denoising cannot remove. In the singular limit the floor is *closed form*: it equals $1/(8\bar{\kappa}^2)$ in the network's slope cap $\bar{\kappa}$, with no dependence on the sampling schedule. A fixed model therefore collapses without bound as the manifold thins, and staying bounded requires capacity growing like $\sqrt{J}$ in the Fisher information J. Empirically the trap is two-horned: the stochastic sampler sticks $\sim 11\times$ too wide, while the deterministic one collapses to a point; neither reaches the data. *Constructively*, we prove that periodically re-matching the model's output to a small, *fixed* pool of real examples makes the loop *memoryless*, so that error stops compounding for every positive fraction of real data. The match rescales the output to the *local width* of the reference; it does not re-insert the reference examples into training. Over a 20-generation head-to-head the prior state of the art plateaus at $9.8\sigma^2$ while anchoring holds $1.01\sigma^2$, and on pixel-space MNIST anchoring removes $\sim$81% of the off-manifold collapse. A preliminary scale-up to CIFAR-10 in raw pixel space (3,072 dimensions), across three seeds, reproduces the separation and, by three independent nearest-neighbor metrics, shows the anchor keeps full coverage of the real data, ruling out mode collapse. The core experiments run on CPU and all are reproducible from released scripts.

1 Introduction

In plain terms. Train an image generator, use it to make pictures, then train the next generator partly on those pictures, and repeat. Each round the model drifts a little further from real data, the way a photocopy of a photocopy loses detail. This is *model collapse*. Recent theory shows the drift can be repaired when the data is "smooth," but leaves open the case that actually describes images, audio, and most real data: points that lie on a thin surface (a *manifold*) inside a much larger space. This paper handles that case. We prove there is a hard floor on how faithful a fixed-size model can stay, set by a single number, how sharp an edge the network can represent, that no amount of tuning removes. Then we give a cheap fix: each round, gently rescale the model's output so its fine-scale spread matches a small, fixed sample of real data. The fix stops the drift from compounding, for any positive amount of fresh real data.

Web-scale corpora already contain machine-generated data, and synthetic data is used deliberately in modern training pipelines. When a generative model is repeatedly retrained on a pool containing its own previous outputs, quality degrades across generations, a phenomenon called *model collapse* [2–5]. For diffusion models, Khelifa et al. [1] give the sharpest analysis to date: with a fraction λ of fresh real data per generation and an annealed truncation schedule $t_0(g) \to 0$, the recursion converges back to the data distribution, *provided the data has finite Fisher information* (their Case 1). They explicitly leave open their Case 2: data supported on, or concentrated near, a low-dimensional manifold, where the Fisher information $J \asymp (d-k)/\sigma^2$ diverges as the noise width $\sigma \to 0$. Essentially all natural data (images, audio, molecular conformations) is of this second kind [6, 7].

This paper answers Case 2.

The idea in one paragraph. When a model repeatedly learns from its own samples, the hardest property to preserve is the data's *thinness*: the exact width of a pen stroke, the sharp edge of a shape. We show that a finite-capacity denoiser can reproduce that thinness only inside a brief window of noise levels, so a small, fixed amount of blur is baked in every generation and accumulates. Two things follow. The blur has an *exact floor* set by a single number, the steepest slope the network can represent, and *no* training or sampling schedule can lower it. And the accumulation can be cancelled by periodically re-matching the model's output to a small, unchanging set of real examples. The match rescales the output to *their local width*; it does not copy the examples back into training. The rest of this section makes both statements precise. The first is our *negative* result (a capacity floor); the second is our *constructive* one (a data-anchoring fix).

The negative half is a *capacity floor* (Section 5). The score that a diffusion model must represent near a σ-thin manifold has normal slope $\kappa(t) = s(t)/\gamma^2(t)$ which is *non-monotone* in diffusion time: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back toward zero as $t \to 0$ (Figure 1a). A network whose normal slope is bounded by $\bar{\kappa}$ therefore under-contracts on a *finite band* of diffusion times; the excess variance frozen in while crossing that band can never be removed afterwards, because too few contraction e-folds remain below the band. We prove that in the singular limit the resulting floor is a universal function of the single ratio $\rho = \bar{\kappa}/\kappa_{\max}$ and admits *closed forms*: under no assumption beyond the slope bound, the floor is exactly $\sigma^2/(2\rho^2) = 1/(8\bar{\kappa}^2)$, so a fixed network collapses relatively without bound as $\sigma \to 0$, and bounded collapse requires $\bar{\kappa} = \Omega(1/\sigma) = \Omega(\sqrt{J})$. No truncation schedule enters these statements: annealing targets a term the schedule can kill, while the floor lives in a term it cannot touch. Empirically the trap is *two-horned* (Figure 2): keeping the reverse-SDE noise, the sampler sticks $\sim$11× too wide; removing it, the recursion collapses to a single point. Neither branch reaches the data.

The constructive half is an anchoring operator (Section 6). A bidirectional, per-axis, local moment match of the training pool (and sampler output) to a small *stale* reference of real data (2,000 samples drawn once at generation zero) pins the normal variance at the reference value each generation. We prove this makes the recursion *memoryless*: the fixed-point equation loses its dependence on the previous generation, the critical fraction disappears, and the dynamics of Khelifa et al. [1]'s Theorem 4.1 are restored in the singular regime. In a 20-generation head-to-head against their annealed-truncation remedy (Figure 5), the annealed arm plateaus at $9.8\sigma^2$ while the anchored arms hold $1.01\sigma^2$, flat, on both samplers we test; on real pixels (MNIST), anchoring closes $\sim$81% of the collapse gap (Figure 6); and a preliminary CIFAR-10 pixel recursion (3,072 dimensions, three seeds) shows the same separation with full mode coverage (Figure 7).

Contributions.

1. **A law of collapse** (Theorem 1): under a measured one-dimensional concave response law, the variance recursion has a unique plateau for $\lambda > \lambda^* = c_\infty/(1+c_\infty)$ and no plateau below; truncation enters only additively, which makes precise why annealing schedules cannot help singular data.

2. **A closed-form capacity floor** (Theorem 2, Corollary 1): any reverse-SDE sampler with normal score slope $\leq \bar{\kappa}$ is floored; in the singular limit the floor is $1/(8\bar{\kappa}^2)$ unconditionally and $\big((1+3e^{-4})/8\big)\bar{\kappa}^{-2}$ for the empirically observed no-overshoot class: two constants that differ by only 5.5%. Bounded relative collapse as $\sigma \to 0$ requires $\bar{\kappa} \geq 1/(2\sigma\sqrt{2C})$.

3. **A memorylessness theorem for anchoring** (Theorem 3): matching pool and output normal variance to a stale real reference removes the recursion's memory, for every $\lambda > 0$, with an error bound independent of the ambient dimension.

4. **Falsifiable empirics**: a zero-fit prediction of the measured sampling floor from one measured network property; a two-horned ablation separating the floor's existence (sampler noise) from its size (slope ceiling); a 20-generation, 5-seed head-to-head win over the state-of-the-art remedy; a pixel-space MNIST recursion; and a preliminary CIFAR-10 pixel-space recursion (three seeds) that rules out mode collapse by three independent nearest-neighbor metrics.

Throughout we keep the epistemic status of each claim explicit: *we prove* (mathematics, given stated assumptions), *we observe* (multi-seed experiments), or *we model* (measured structural laws that we do not derive). Section 8 collects everything in the third category.

2 Related work

Model collapse. Collapse under recursive training is documented and analyzed across model families [2–5, 8]; accumulation (rather than replacement) of data and sufficient fresh-data fractions mitigate it in the absolutely continuous setting. Khelifa et al. [9] bound error propagation in recursively trained diffusions via a discounted accumulation argument, but their assumptions require absolutely continuous data (structurally excluding manifolds), with fixed truncation and upper bounds rather than laws. Khelifa et al. [1] characterize the limiting collapse distribution spectrally and prove annealed truncation converges for finite-Fisher data (their Case 1), stating manifold-supported data (Case 2) as open. We give the Case 2 answer: a floor theorem for their sampler class, and an operator restoring their Case 1 dynamics.

Diffusion on manifold data. That diffusion models detect and concentrate on data manifolds is established [7, 10]. The singular ($\sigma \to 0$) score is the object of a growing one-shot literature: optimal denoising under low regularity [11], provable learning under the manifold hypothesis [12], and mitigation of the score singularity [13]. These works are one-generation; our subject is the *recursion*, where a per-generation floor compounds into a permanently displaced plateau. We use the Tweedie/posterior-mean jump [11, 14] as a raw-error reducer inside the loop and are explicit that it does not, by itself, break the floor (Section 6).

Local geometry estimation. Our anchoring operator estimates local tangent frames by kNN-PCA; its error analysis imports standard local-PCA perturbation rates [15, 16]. The operator needs only *shared* frames between pool and reference, not correct ones, which is why these rates enter only at second order (Lemma 5).

3 Setup

Data model (σ-tube). Let $M \subset \mathbb{R}^d$ be a compact smooth k-dimensional manifold with reach $\tau > 0$. Data are $X_0 = \varphi(U) + \sigma N_\perp$, with U uniform on M and $N_\perp$ standard normal in the $(d-k)$-dimensional normal space; $\sigma > 0$ is the tube width. The singular regime is $\sigma \to 0$; the Fisher information of the marginal scales as $J \asymp (d-k)/\sigma^2 \to \infty$ (Case 2 of 1).

Forward process and the normal coordinate. We use the variance-preserving OU forward process $X_t = a(t)X_0 + s(t)Z$ with $a(t) = e^{-t/2}$, $s^2(t) = 1 - e^{-t}$ [17, 18]. Fix a point of M and work in a normal coordinate $x \in \mathbb{R}$, treating the fiber marginal as Gaussian and decoupled from tangential position (*flat-tube idealization*; curvature enters at $O(s^2/\tau)$ and is tracked where it matters). Then $X_0 \sim \mathcal{N}(0, \sigma^2)$, $X_t \sim \mathcal{N}(0, \gamma^2(t))$ with $\gamma^2 = a^2\sigma^2 + s^2$, and the denoising target is linear:

$$\varepsilon^*(x,t) = \kappa(t)\, x, \qquad \kappa(t) = \frac{s(t)}{\gamma^2(t)}, \qquad \nabla \log p_t(x) = -\frac{x}{\gamma^2(t)}. \tag{1}$$

κ is the normal slope the score network must represent. It satisfies $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$, and falls back to 0 as $t \to 0$ (Figure 1a). In the singular limit the *peak* slope diverges, which is the precise sense in which infinite Fisher information bites the estimator.

Samplers. $S_{\text{trunc}}(t_0)$ integrates the learned reverse SDE from T down to $t_0 > 0$ (Euler-Maruyama, $x \leftarrow x + (-\frac{1}{2}x + \hat{\varepsilon}/s)\,dt + \sqrt{|dt|}\,\xi$) and outputs x_{t_0}; all prior remedies, including the annealed schedules of Khelifa et al. [1], are of this class. $S_{\text{jump}}(\hat{t})$ integrates only to a moderate $\hat{t}$ and then outputs the Tweedie posterior-mean jump $\hat{x}_0 = (x_{\hat{t}} - s(\hat{t})\,\hat{\varepsilon})/a(\hat{t})$ [11, 14].

The recursion. For a distribution p on $\mathbb{R}^d$ define the per-direction normal second moment $v(p) = \mathbb{E}_p[\text{dist}(X, M)^2]/(d-k)$; true data has $v = \sigma^2$. Generation g trains $\hat{\varepsilon}_g$ by denoising score matching on $\text{pool}_g = \lambda\,(\text{fresh real}) + (1-\lambda)\,(\text{generation } g{-}1)$ and samples; v_g is the output value. Since v is linear in the measure, $w_g := v(\text{pool}_g) = \lambda\sigma^2 + (1-\lambda)v_{g-1}$ exactly.

Modeling assumptions (*we model*). Our law rests on two assumptions about the trained pipeline that we *measure and cross-validate* but do not derive.

Assumption 1 (scalar reduction)**.** *The per-generation map factors through the pool width: $v_g = F_g(w_g)$.* Why a scalar: *the collapse observable is a* variance *along the normal fibre, one number per normal axis by construction, so this is an order-parameter statement (the width closes on itself) rather than a claim that the network is scalar, in the same way an order parameter closes a mean-field recursion without the microscopic state being low-dimensional.* Validation: *fixed points measured in one experiment type predict those of an independent type within* 1 *to* 13% *across six* (t_0, λ) *cells (Section 7).*

Assumption 2 (concave response)**.** $F_g(w) = a^2(t_0(g))\, w + s^2(t_0(g)) + e^2(w)$*, where the sampling excess* $e^2 : [0,\infty) \to (0,\infty)$ *is nondecreasing and concave with* $e^2(0) = e_0^2 > 0$*.* Validation: *the measured local slope of* e^2 *runs from* $+0.6$ *(fat pools) to* ≈ 0 *(thin pools), concave and saturating; its small magnitude is itself a measured near-cancellation of two* $O(0.7)$ *channels (Section 7).*

4 The law of collapse

Under Assumptions 1 and 2 the recursion is one-dimensional:

$$v_g = T(v_{g-1}) + s_g^2, \qquad T(v) = \lambda\sigma^2 + (1-\lambda)v + e^2\big(\lambda\sigma^2 + (1-\lambda)v\big), \tag{2}$$

with $s_g^2 := s^2(t_0(g))$ the truncation remainder. Let $c_\infty := \lim_{w\to\infty} e^{2\prime}(w)$ (exists by concavity) and define the *critical real-data fraction* $\lambda^* := c_\infty/(1+c_\infty)$.

Theorem 1 (law of the collapse; *we prove*)**.** *Fix $\lambda \in (0,1]$.* ***(i)*** *If $\lambda > \lambda^*$, T has a unique fixed point $v_\infty > 0$ solving $v = \lambda\sigma^2 + (1-\lambda)v + e^2(\lambda\sigma^2 + (1-\lambda)v)$, and for every $v_0 \ge 0$ and every truncation schedule with $s_g^2 \to 0$ the iterates converge, $v_g \to v_\infty$, with local rate $\rho_{\text{loc}} = (1-\lambda)\big(1 + e^{2\prime}(w_\infty)\big)$.* ***(ii)*** *If $(1-\lambda)(1+c_\infty) > 1$, then $T(v) - v \ge T(0) > 0$ for all v and $v_g \to \infty$: no plateau, for any schedule.* ***(iii)*** *In the affine special case $e^2(w) = e_0^2 + cw$: $\lambda^* = c/(1+c)$, $\rho_{\text{loc}} = (1+c)(1-\lambda)$, and $v_\infty = \big((1+c)\lambda\sigma^2 + e_0^2\big)/\big(\lambda(1+c) - c\big)$.*

Proof. Appendix A. Elementary but complete: concavity gives a unique downcrossing of $T(v) - v$; contraction above the fixed point plus a monotone barrier below give convergence under vanishing perturbations. □

Remark 1 (why annealing cannot help singular data). Truncation enters (2) only through the additive, schedule-killable s_g^2. The plateau v_∞ is determined by e^2 alone, which no schedule touches. The remedy of Khelifa et al. [1] is the special case $e^2 \equiv 0$ (their Case 1 assumptions make the estimation term schedule-dominated), where indeed $v_\infty = \sigma^2$. Case 2 is precisely the regime where e^2 has a floor, the subject of the next section.

5 The capacity floor

We now prove that for manifold data the sampling excess e^2 of any truncated reverse-SDE sampler is floored by network capacity. Write the trained ε-predictor's affine part as $\hat\varepsilon(x,t) = \hat\kappa(t)x + b(t) + \delta(x,t)$, where $(\hat\kappa, b)$ is the affine $L^2(p_t)$ fit of the estimator itself and δ the orthogonal residual. Two structural facts (proofs in Appendix B):

Lemma 1 (variance ODE; *we prove*)**.** *Under the reverse Euler scheme with affine estimator, the ensemble normal variance obeys, in the small-step limit,*

$$-\frac{dV}{dt} = \Big(1 - \frac{2\hat\kappa(t)}{s(t)}\Big)V + 1. \tag{3}$$

With the exact slope $\hat\kappa = \kappa$, $V(t) = \gamma^2(t)$ solves (3) exactly. The forcing $+1$ is the sampler's own injected noise.

Lemma 2 (orthogonality shield; *we prove*, unconditional)**.** *For any estimator, the residual δ satisfies $\mathbb{E}_{p_t}[\delta] = \mathbb{E}_{p_t}[x\delta] = 0$ (normal equations), hence has no first-order effect on (3): at first order the residual can only* add *variance. Thinning can enter only through the affine part.*

By Lemma 2, lower bounds derived from (3) with the affine part alone bound the full estimator. This is the precise sense in which the floor is *not* an artifact of linearizing the score: a real score network is highly nonlinear, but its nonlinearity is carried entirely by the orthogonal residual δ, which by the normal equations has zero first-order effect on the variance and can only *raise* it, so the affine lower bound is a lower bound for the fully nonlinear network. The crux is the geometry of the required slope:

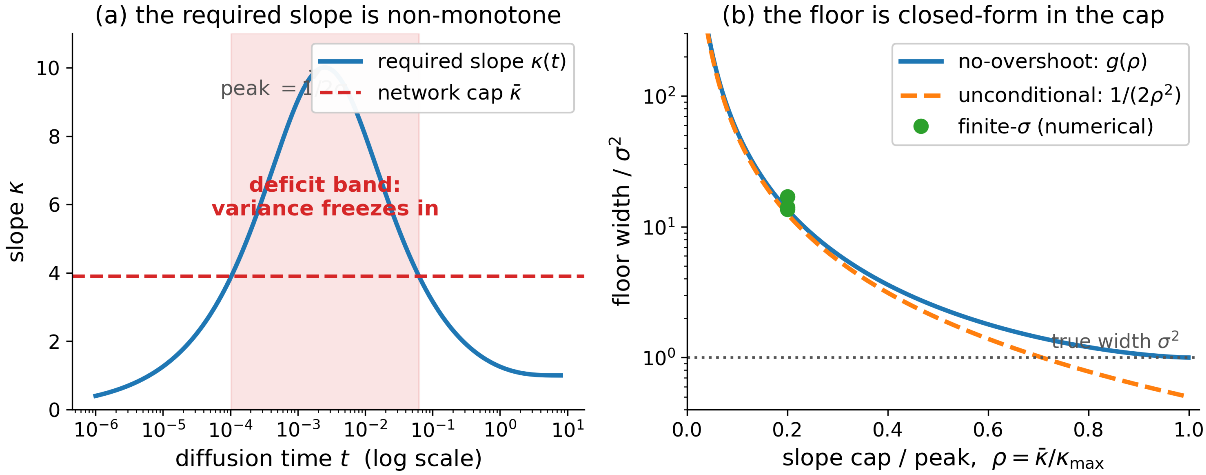

Figure 1: **The mechanism and the law.** (a) The required normal slope $\kappa(t) = s/\gamma^2$ is non-monotone: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back to zero. A network with ceiling $\bar{\kappa} < \kappa_{\max}$ under-contracts only on a *finite band* (shaded); variance frozen in the band cannot be removed below it (too few contraction e-folds remain). (b) The resulting floor in units of σ^2 as a function of $\rho = 2\sigma\bar{\kappa}$: closed-form no-overshoot law $g(\rho)$ (blue, Theorem 2), exact unconditional law $1/(2\rho^2)$ (orange), finite-σ ODE values converging to the limit (green markers at $\rho = 0.2$).

Lemma 3 (finite band; *we prove*)**.** *$\kappa(t)$ rises from 0, peaks at $\kappa_{\max} = 1/(2\sigma)$ at $t \approx \sigma^2$, and falls back to 0 as $t \to 0$. Hence for any ceiling $\bar{\kappa} < \kappa_{\max}$ the deficit set $\{t : \kappa(t) > \bar{\kappa}\}$ is a finite band; below it the exact slope is representable again, but only $2\log(1 + \rho^2/4) \le \rho^2/2$ contraction e-folds remain ($\rho := 2\sigma\bar{\kappa}$), so variance frozen in the band persists to $t_0 \to 0$.*

In the singular limit the whole flow collapses onto a one-parameter family (Appendix C for proofs and error terms):

Lemma 4 (singular-limit ODE; *we prove*)**.** *Rescale $t = \sigma^2 u$, $V = \sigma^2 W$, and write the slope envelope as $\hat{\kappa}(t) = (1/\sigma)m(u)$ with $m(u) = \rho/2$ (unconditional envelope) or $m(u) = \min\{\sqrt{u}/(1+u), \rho/2\}$ (no-overshoot envelope). Then (3) becomes, as $\sigma \to 0$ at fixed ρ,*

$$-\frac{dW}{du} = 1 - \frac{2m(u)}{\sqrt{u}}\,W, \tag{4}$$

*a σ-free equation; the uncapped equation has exact solution $W = 1 + u$ (the rescaled true marginal), and deviations contract quadratically under downward integration. Consequently $\lim_{t_0 \to 0} V(t_0)/\sigma^2 = g_{\text{class}}(\rho) + O(\sigma^2/\rho^2)$: **universality in ρ is a theorem**.*

Theorem 2 (closed-form capacity floor; *we prove*)**.** *Let the trained score's normal slope satisfy $\hat{\kappa}(t) \le \bar{\kappa} < \infty$ (finite normal Lipschitz constant, automatic for any finite network), and set $\rho = 2\sigma\bar{\kappa}$. Then for every truncation time $t_0 > 0$ and every schedule:*

(U) Unconditional. *With no further assumption (arbitrary overshoot permitted),*

$$V(t_0) \;\ge\; \sigma^2\,\frac{1}{2\rho^2}\,(1 + o(1)) \;=\; \frac{1}{8\bar{\kappa}^2}\,(1 + o(1)) \qquad (\sigma \to 0), \tag{5}$$

where $1/(2\rho^2)$ is the exact *value of (4) with $m \equiv \rho/2$: $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}}\,du = 1/(2\rho^2)$. The floor exceeds σ^2 whenever $\rho < 1/\sqrt{2}$, i.e. $\bar{\kappa} < \kappa_{\max}/\sqrt{2}$.*

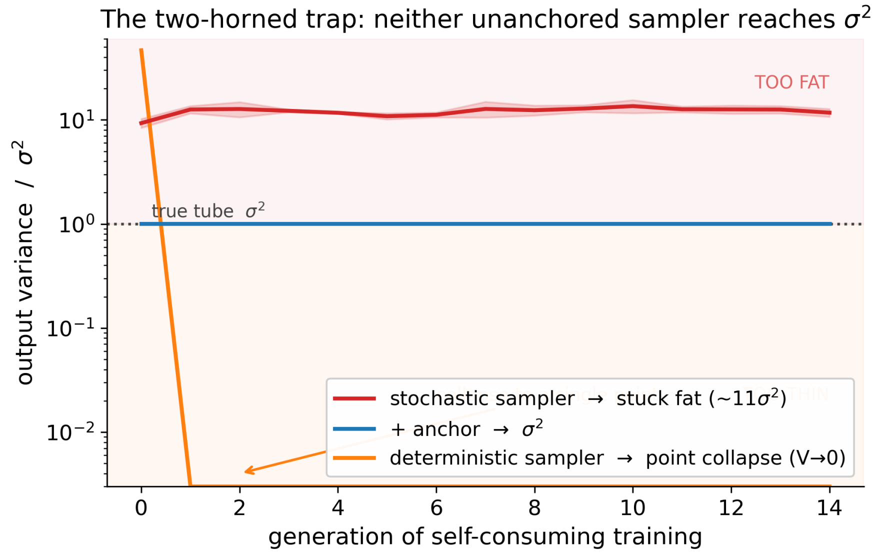

Figure 2: **The two-horned trap (*we observe*).** Self-consuming recursions at fixed $t_0 = 10^{-4}$, no anchor, 3 seeds. With the reverse-SDE noise on, the recursion sticks fat at 10.8 to 13.2 σ^2 (red). With the noise zeroed (deterministic sampler), it collapses to a *single point* within one generation (orange; verified loss of all 2-D spread, no numerical overflow). Neither branch reaches the tube (σ^2, dotted); the anchored recursion (blue, Section 6) holds it.

(N) No-overshoot. *If additionally $\hat\kappa(t) \le \kappa(t)$ (the empirically observed profile; the network never over-contracts), the tighter floor $V(t_0) \ge \sigma^2\, g(\rho)(1+o(1))$ holds with the explicit closed form: writing $q = \sqrt{1-\rho^2}$, $x_\mp = (1 \mp q)/\rho$, $u_\mp = x_\mp^2$,*

$$W_{\text{exit}} = (1+u_+)e^{-4q} + \frac{(3-2q)-(3+2q)e^{-4q}}{2\rho^2}, \qquad g(\rho) = 1 + \frac{W_{\text{exit}} - (1+u_-)}{(1+u_-)^2}. \tag{6}$$

$g(\rho) > 1$ for every $\rho \in (0,1)$ with $g \to 1$ only as $\rho \to 1^-$, and $g(\rho) = C_0/\rho^2 + O(1)$ as $\rho \to 0$ with $C_0 = (1+3e^{-4})/2 = 0.5275$, i.e. $\Phi_{\text{det}} \to \big((1+3e^{-4})/8\big)\bar\kappa^{-2} = 0.1319\,\bar\kappa^{-2}$.

The two asymptotic constants, $1/8$ and $(1+3e^{-4})/8$, differ by 5.5%: the class distinction is quantitatively minor. The closed forms match direct integration of (4) to $\le 2.4 \times 10^{-5}$ across $\rho \in [0.1, 1]$, and the finite-σ ODE converges to them from above at the stated $O(\sigma^2/\rho^2)$ rate.

Corollary 1 ($\sigma \to 0$: the answer to Case 2; *we prove*)**.** *Hold $\bar\kappa$ fixed and let $\sigma \to 0$ ($J \to \infty$). Then unconditionally $V/\sigma^2 \ge (1+o(1))/(8\bar\kappa^2\sigma^2) \to \infty$: a fixed-capacity truncated sampler collapses* relatively *without bound, for every annealing schedule. Conversely, bounded relative collapse $V/\sigma^2 \le C$ requires*

$$\bar\kappa \;\ge\; \frac{1}{2\sigma\sqrt{2C}} \;=\; \Omega(1/\sigma) \;=\; \Omega(\sqrt{J}). \tag{7}$$

Truncated sampling of singular data demands capacity growing with the singularity; no fixed architecture provides it.

What each ingredient does (*we observe*). The ablation of Figure 2 separates the roles exactly as (3) dictates: the sampler's own noise (the +1) is load-bearing for the floor's *existence*: switching

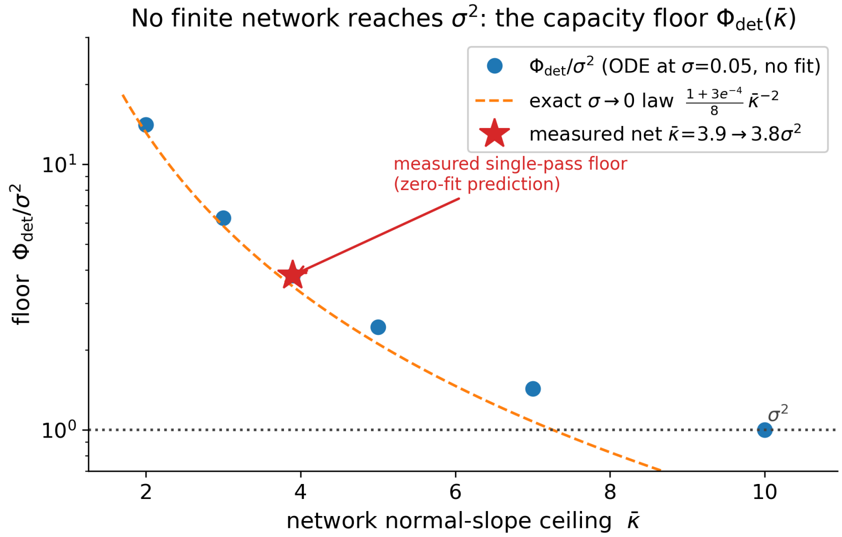

Figure 3: **The capacity floor, predicted and measured.** Blue: finite-σ variance-ODE floor at $\sigma = 0.05$ (no fit). Dashed: the exact $\sigma \to 0$ law $\big((1{+}3e^{-4})/8\big)\bar\kappa^{-2}$ of Theorem 2. Star: the measured network ceiling $\bar\kappa = 3.9$ predicts the measured sampling floor $3.8\sigma^2$ with zero free parameters.

it off sends $V \to 0$ (point collapse, the deterministic horn), while the finite ceiling $\bar\kappa$ makes the floor *larger than* σ^2 (the stochastic horn). The two branches bracket the target from both sides; no choice of the sampler noise magnitude attains it.

Zero-fit confrontation with a real network (*we observe*). For our standard pipeline the measured slope profile tracks $\kappa(t)$ faithfully at benign times ($\hat\kappa/\kappa = 0.95$ to 0.97 for $t \geq 0.1$) and then saturates at $\bar\kappa \approx 3.9$ while κ continues to $\sim$10, an optimization ceiling independent of sample size. Feeding $\bar\kappa = 3.9$ ($\rho = 0.39$) into the floor law predicts $\Phi_{\text{det}} = 3.8\,\sigma^2$ at $\sigma = 0.05$, matching the same network's measured $t_0 \to 0$ sampling floor with *no free parameter* (Figure 3); integrating (3) with the full measured profile $\hat\kappa(t)$ reproduces 69 to 75% of the measured floor at each of three truncation times, with the exact-score control reproducing $\gamma^2(t_0)$ to all displayed digits. The remaining $\sim$30% is the residual-field channel of Lemma 2, which we bound in a block-correlation model (*we model*; Appendix E).

An honest gap, half closed (*we model*). Iterating the floor through the recursion (every generation's pool is at least as wide as $\lambda\sigma^2 + (1-\lambda)\Phi_{\text{det}}$) with the network's *clean-tube* ceiling $\bar\kappa = 3.9$ gives a deterministic fixed point $v^*_{\text{det}} = 4.5\,\sigma^2$. The measured recursion floor is $\approx 11\,\sigma^2$, a factor 2.4 above; our theorem claims only a *lower bound* on the recursion (respected). The single-width probe (Appendix F) accounts for roughly half of the excess: the ceiling itself degrades on fatter training pools — *we observe* $\bar\kappa(w) = 4.43 - 11.6\sqrt{w}$ across $\sqrt{w} \in [0.03, 0.13]$ (max fit residual 0.02) — so a network trained on self-generated, fattened data has a worse slope profile than the clean-tube analysis assumes. Feeding this measured law back self-consistently, $v \leftarrow \Phi_{\text{det}}\big(w(v), \bar\kappa(w(v))\big)$, moves the fixed point to $6.1\,\sigma^2$ at $t_0 = 10^{-4}$ ($7.4\,\sigma^2$ at 10^{-3}) with no new parameters, and the resulting pool width $\sqrt{w^*} = 0.094$ lies *inside* the measured support of the $\bar\kappa(w)$ law (interpolation, not extrapolation). The remaining factor ≈ 1.5–1.8 we attribute to stochastic per-generation compounding that no deterministic pass captures; its constant stays open.

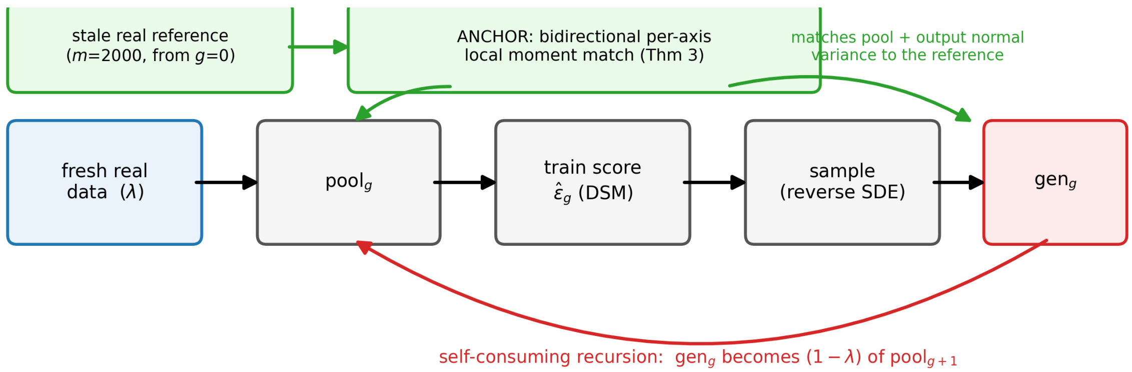

Figure 4: **Where the anchor sits.** Each generation, the pool (and, for the jump pipeline, the sampler output) is passed through a bidirectional per-axis local moment match against a *stale* reference of $m = 2{,}000$ real samples drawn once at $g = 0$. No provenance labels are needed; the reference is never refreshed.

6 The anchoring fix

Corollary 1 says the singular regime cannot be won by scaling the model or tuning the schedule. It can be won by *anchoring the recursion to data*.

The operator. For each point, find its k nearest neighbors in the reference; take the local PCA frame (top-k_{dim} directions tangent, remainder normal); rescale the point's normal coordinates so that the pool's local normal second moments equal the *reference's* local normal second moments in the *same estimated frames*. The match is *bidirectional* (it contracts fat directions and inflates thin ones) and *per-axis*. Both properties are forced by the theory: the sampler's error is signed (fat from injection, thin from posterior-mean sharpening; Proposition 1 in Appendix D) and anisotropic whenever the normal spectrum straddles $s^2(\hat{t})$, so any scalar or one-sided matcher mis-corrects some directions (*we observe*: a scalar matcher leaves MNIST-latent thin ratios at 0.62; the per-axis matcher achieves $|\text{thin} - 1| \leq 0.012$).

Lemma 5 (shared-frame cancellation; *we prove* to first order)**.** *Let both reference and pool normal energies be measured in the same kNN-PCA frames estimated from the reference. Then the matched pool satisfies, per normal direction,* $v(\text{matched}) = \hat{\sigma}^2_{\text{ref}} + O_p(m^{-1/2}) + \Delta_{\text{leak}}$*, where the frame-error leakage* Δ_{leak} *afflicts pool and reference* identically *at first order in the frame perturbation and cancels; standard local-PCA rates [15, 16] enter only at second order. Every quantity in the bound is intrinsic* (k, τ, σ, m)*; the ambient dimension* d *appears nowhere.*

Theorem 3 (anchoring makes the recursion memoryless; *we prove* given Lemma 5 and Assumption 1)**.** *Insert the matcher after pooling (and after sampling, for the jump pipeline). Then the training input has normal variance* $\hat{\sigma}^2_{\text{ref}} + O(\text{err})$ regardless of v_{g-1}*: the map (2) loses its* v*-argument,*

$$v_g = F_g\big(\hat{\sigma}^2_{\text{ref}} + O(\text{err})\big),$$

a constant sequence up to non-compounding $O(\text{err})$*. Consequently: no fixed-point equation,* ***no critical fraction, and stability for every*** $\lambda > 0$*; and with* S_{trunc} *the dynamics reduce to truncation-only, restoring the hypotheses and conclusion of Khelifa et al. [1]'s Theorem 4.1 in the singular regime.*

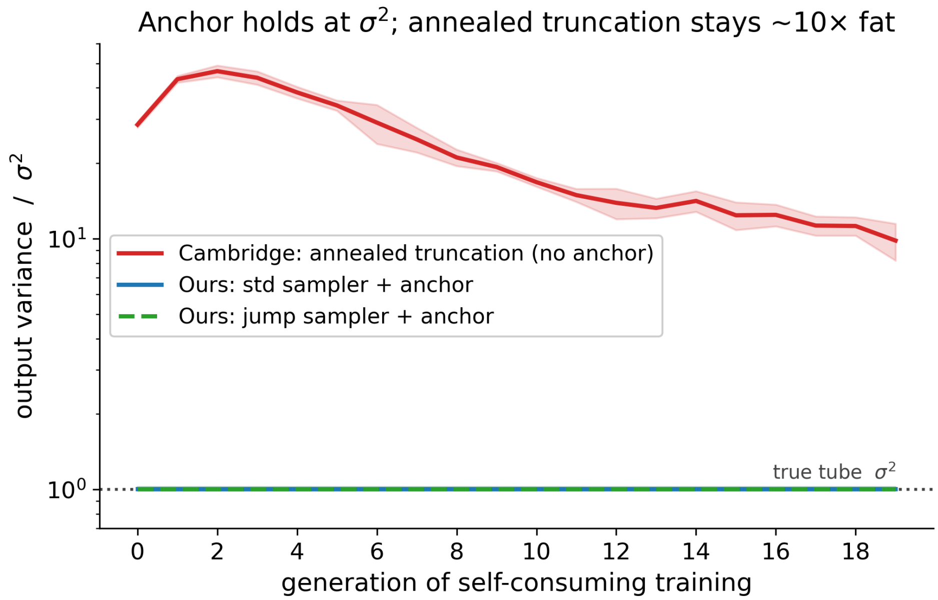

Figure 5: **Head-to-head vs. the state-of-the-art remedy (*we observe*).** Ring manifold, $\sigma = 0.05$, $\lambda = \frac{1}{2}$, 20 generations, 5 seeds (mean ± s.d.). Red: annealed truncation $t_0 : 0.05 \to 10^{-4}$, no anchor [1]; it rises to $47\sigma^2$, then thins only to $9.8\sigma^2 \pm 1.7$. Blue/green: the same recursion with the anchor (standard and jump samplers): $1.01\sigma^2 \pm 0.02$, flat for 20 generations.

Why this is not a replay buffer. Replay, data-accumulation, and data-refresh remedies [4, 8] fight collapse by *re-inserting real examples* into the pool: they preserve *samples*. The anchor preserves something weaker and more portable, the *local normal moments* (per-axis widths) of a small stale reference, and imposes them on *whatever the model currently generates*. It never adds a reference point to the training pool, uses no provenance labels, and reuses the same m=2,000 points for every generation. The distinction is not cosmetic. A replay buffer of size m still lets the $(1-\lambda)$ synthetic majority set the pool's fine-scale statistics, so the floor of Section 5 still governs that majority; anchoring instead *overwrites* the fine-scale statistics of the majority itself, which is why it removes the floor rather than diluting it. In the language of the law (Theorem 1): replay changes λ; anchoring changes the response map F_g.

Attribution, stated plainly. In our pipeline the *matcher* is the floor-breaker; the Tweedie jump is a *raw-error reducer*. The jump's raw output is itself fat (7 to $20 \times \sigma^2$ across settings); its value is a $\approx 2\times$ smaller one-shot raw error and $\approx 3\times$ smaller unanchored plateau (0.052 vs 0.17 at $\lambda = \frac{1}{2}$, itself predicted by Theorem 1 as $e^2_{\text{jump}}/\lambda$), plus smaller post-match tails. Post-anchor, the standard and jump samplers converge to the same variance; we report both.

Dimension scaling (*we observe*). Lemma 5's d-independence is a postdiction: across ambient $d = 8 \to 128$ (curved 2-manifolds, fixed tube), the *raw* one-shot floors grow near-linearly ($\sim d^{0.76}$ standard, $\sim d^{0.95}$ jump, the collapse itself), while the *matched* floors are flat ($\sim d^{0.1}$), and anchored recursions are flat for 8 generations at both $d = 8$ and $d = 128$.

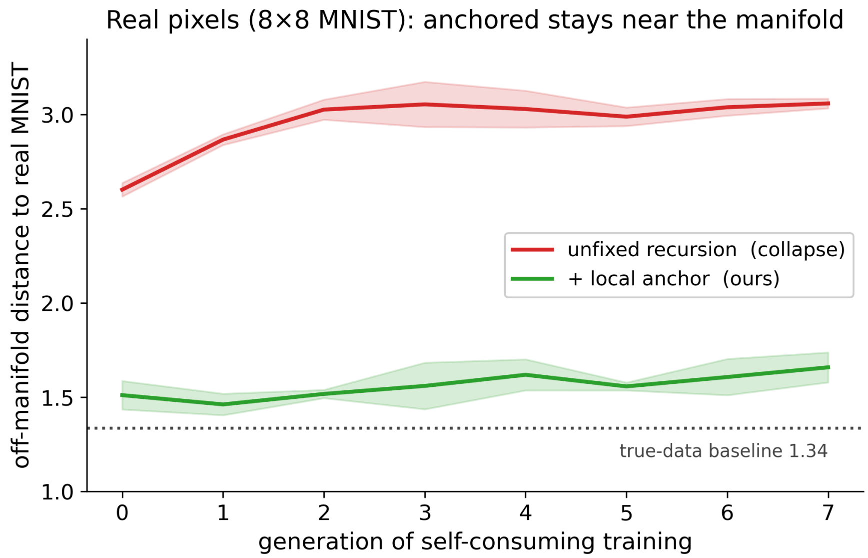

Figure 6: **Real pixels (*we observe*).** Self-consuming recursion on 8×8 MNIST (64-dimensional pixel space), $\lambda = \frac{1}{2}$, 8 generations, 3 seeds. Off-manifold distance = mean nearest-neighbor distance to a held-out real test set. Unanchored drifts to 3.06 (2.3× the true-data baseline 1.34); anchored holds 1.66 (1.24×), closing ∼81% of the gap. The mild upward creep of the anchored arm reflects the local-chart approximation on a genuinely curved manifold.

7 Experiments

All experiments are CPU-only (≤ 12 threads), use small MLPs (3 hidden layers, width 256, SiLU, sinusoidal time embedding), denoising score matching, and geometric reverse-time grids; every long run streams to an append-only log and is exactly resumable. Full protocols, seeds, and a script-to-claim map are in the released code (Section 9).

Protocols. *Ring*: radius 2.5, $\sigma = 0.05$ ($\sigma^2 = 0.0025$), 4,000 samples/generation, 2,000 training steps. *Head-to-head* (Figure 5): $G = 20$, $\lambda = \frac{1}{2}$, 5 seeds; arm A anneals $t_0 : 0.05 \to 10^{-4}$ geometrically (the remedy of 1); arms B/C add the anchor to the standard/jump samplers. *Two-horned ablation* (Figure 2): fixed $t_0 = 10^{-4}$, noise on/zeroed, $G = 15$, 3 seeds. *Pixel MNIST* (Figure 6): 8×8 pixels, $G = 8$, 3 seeds, local matcher with $k = 96$, $k_{\mathrm{dim}} = 12$. *Law identification*: fixed-t_0 recursions at $t_0 \in \{0.02, 0.05\}$, $\lambda \in \{0.25, 0.5, 0.75\}$; the affine law fitted at one t_0 predicts fixed points at the other within 1 to 13% (Assumption 1 validation), and the measured excess-response slope runs $+0.6 \to\approx 0$ across pool widths (Assumption 2 validation), the net of a measured deficit channel -0.68 and injection channel $+0.73$; a refinement using the probe's own capacity measurements splits the latter further (Appendix F).

Scaling to CIFAR-10 pixels (*we observe*). To test whether the mechanism survives at realistic dimension, we ran the same self-consuming recursion in raw CIFAR-10 pixel space ($32 \times 32 \times 3 =$ 3,072 dimensions) with a small convolutional denoiser (∼7M parameters, mixed precision), $\lambda = \frac{1}{2}$, $G = 8$, three seeds (Figure 7a). The unanchored recursion settles into a worse, noisier state (off-manifold distance 22.8 ± 1.2 vs. the true-data baseline 20.1) while the anchor holds a stable plateau (16.3 ± 0.7); the gap has the same sign in every seed ($+6.5 \pm 1.1$). Because a low off-manifold distance could in principle hide mode collapse (a few good samples repeated), we check three

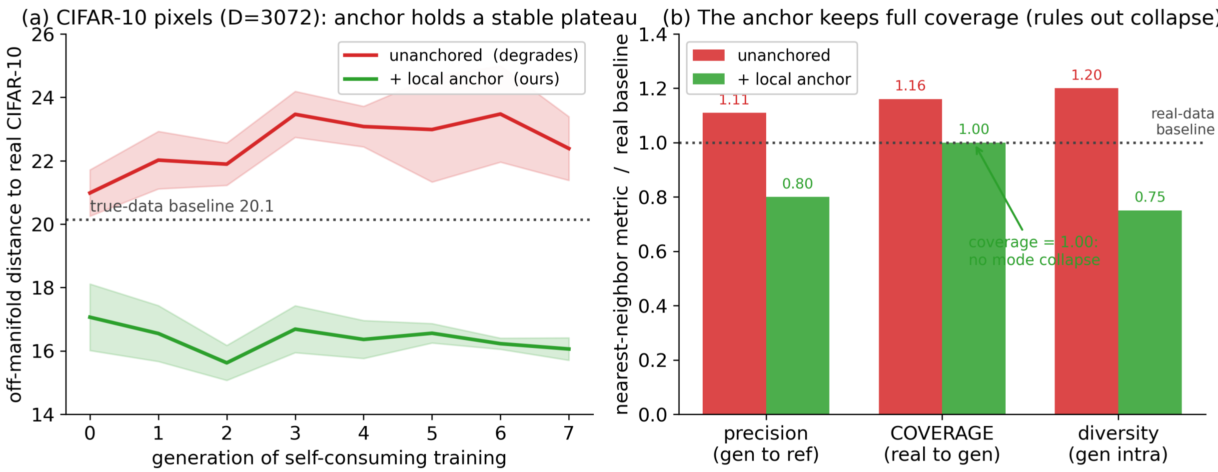

Figure 7: **CIFAR-10 pixel recursion (*we observe*).** (a) Off-manifold distance (mean nearest-neighbor distance to held-out real CIFAR-10) across 8 self-consuming generations, 3 seeds (mean ± s.d.). The anchor holds a stable plateau well below the unanchored arm, which settles into a worse, noisier state; the separation holds in every seed. (b) Three nearest-neighbor metrics relative to a real-vs-real baseline (dotted). The anchor keeps *coverage* of the real data at 1.00× (no real modes dropped, ruling out mode collapse); its 0.75× diversity is a bounded, non-compounding contraction cost (identical at generations 4 and 7).

independent nearest-neighbor metrics against a real-vs-real baseline (Figure 7b): the anchored arm keeps *coverage* of the real data at 1.00× (the unanchored arm loses coverage, 1.16×), so no real modes are dropped, and its reduced intra-set diversity (0.75×) is a bounded, non-compounding cost of the match. We are careful about what this shows. At this model budget the samples are blurry; this is a check on the collapse *dynamics* and a ruling-out of mode collapse, not a demonstration of high perceptual quality, and the sub-baseline off-manifold distance reflects mild contraction toward the manifold rather than superiority to real data. The unanchored arm reaches a stable displaced fixed point rather than runaway collapse, exactly as Theorem 1 predicts at $\lambda = \frac{1}{2}$, where fresh-data injection pins a finite plateau.

8 Limitations and scope

Everything in the *we model* tier is collected here. (1) Assumptions 1 and 2 are measured, cross-validated laws about the trained pipeline, not derived facts; we present them as such, in the spirit of measured critical exponents. (2) The recursion floor constant is partially open: our closed forms are exact for a single pass, and lower bounds for the recursion; the measured recursion floor sits a factor 2.4 above the iterated single-pass bound, of which roughly half is accounted for by feeding the measured capacity-degradation law $\bar{\kappa}(w)$ back self-consistently (Section 5), leaving a 1.5–1.8× stochastic-compounding residual unmodeled. (3) The residual-field ($\sim 30\%$ single-pass) channel is bounded in a block-correlation model (Appendix E); negative association across blocks is the one direction that could weaken that particular bound (the closed-form floor of Theorem 2 does not depend on it). (4) Curvature is carried at order $O(s^2/\tau)$ in the flat-tube idealization; sphere experiments bound the resulting bias empirically (non-compounding), but we do not compute the constants. (5) The anchor controls normal *second moments*; tails and tangential statistics improve

Table 1: Headline numbers (mean ± s.d. over seeds; variance in units of σ^2 except MNIST and CIFAR-10, which report off-manifold distance).

Experiment	Arm	Final value	Seeds
Head-to-head, G=20	annealed truncation [1]	9.84 ± 1.7	5
	standard sampler + anchor	1.01 ± 0.02	5
	jump sampler + anchor	1.01 ± 0.02	5
Ablation, fixed t_0=10^{-4}	stochastic (noise on)	10.8 to 13.2	3
	deterministic (noise zeroed)	0 (point collapse)	3
Pixel MNIST, G=8	unanchored	3.06	3
	anchored	1.66	3
	true-data baseline	1.34	n/a
Pixel CIFAR-10, G=8	unanchored	22.8 ± 1.2	3
	anchored	16.3 ± 0.7	3
	true-data baseline	20.1	n/a
	anchored coverage / diversity	1.00× / 0.75×	3
Floor prediction	measured $\bar{\kappa}$=3.9 $\Rightarrow \Phi_{\rm det}$	3.8 (zero fit)	n/a

but are not anchored; on strongly curved data the anchored recursion shows a mild residual creep (Figure 6). (6) A parameter-free one-step prediction of MNIST-latent thin ratios fails quantitatively (it predicts the direction and the regime boundary, not the number); we flag it rather than fit it. (7) Our evidence spans a ring, curved 2-manifolds swept across ambient dimension $d = 8 \to 128$, six (t_0, λ) law-identification cells, the two-horned ablation, pixel-space MNIST, and a preliminary CIFAR-10 pixel-space recursion (3,072 dimensions, three seeds), all chosen so that every quantity the theory names is directly measurable. The CIFAR-10 run confirms the collapse *dynamics* and rules out mode collapse at scale (Figure 7), but at this model budget the samples are blurry and we do not claim a perceptual-quality result; we do not test latent-diffusion pipelines or larger models. The theory predicts the mechanism transfers, because the floor depends only on the *local* normal slope and not on the ambient dimension (Lemma 5, and the measured d-independence of the matched floors); confirming this in a large latent-diffusion system is the clearest next step and is left as future work.

9 Reproducibility

Code, result logs, and figures are public at

`https://github.com/Naman84789/model-collapse-manifolds`

Every claim maps to a script, an append-only result log, and a figure in the released code (ring, ablation, MNIST, floor law, law identification; ∼2 CPU-hours total). The four result figures regenerate from logged data in seconds. A README lists the run order, the load-bearing scripts, and the headline numbers a reproduction must match. The released code also includes an adversarial verification script for Theorem 2: it re-derives the band crossing symbolically (independent of the manuscript's derivation), integrates the limit ODE with an independent solver, confirms the limiting cases $\rho \to 0$, $\rho \to 1^-$, $\bar{\kappa} \to \infty$, and $\lambda \to 1$, verifies that the boundary layer contracts at the predicted quadratic rate, and attacks the comparison bound with 200 randomized slope profiles including overshooting ones; all checks pass.

Acknowledgments

Numerical experiments, figure preparation, and portions of the derivations were carried out with the assistance of an AI system (Claude, Anthropic), used as a research tool under the author's direction; the author reviewed all mathematics and takes full responsibility for every claim.

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A Proof of Theorem 1

Write $g(v) := T(v) - v$ with T as in (2); g is concave (composition of an affine map with the concave nondecreasing $w \mapsto w + e^2(w)$, minus a linear term), with $g(0) = T(0) = \lambda\sigma^2 + e^2(\lambda\sigma^2) > 0$ and one-sided derivative $g'(v) = (1-\lambda)\big(1+e^{2\prime}(w(v))\big) - 1$, nonincreasing in v with limit $(1-\lambda)(1+c_\infty) - 1 < 0$ when $\lambda > \lambda^*$.

(i) Existence, uniqueness. A concave function positive at 0 whose derivative is eventually $\leq -\delta < 0$ tends to $-\infty$, so g has a zero; concavity plus $g(0) > 0$ forbids two downcrossings, so the zero v_∞ is unique, with $T(v) > v$ for $v < v_\infty$ and $T(v) < v$ for $v > v_\infty$.

Contraction above. For $v \geq v_\infty$, monotonicity and concavity give $0 \leq T(v) - v_\infty \leq \rho_{\text{loc}}(v - v_\infty)$ with $\rho_{\text{loc}} = T'(v_\infty^+) < 1$.

Convergence under vanishing perturbations. Let $v_g = T(v_{g-1}) + s_g^2$, $s_g^2 \to 0$. While iterates stay $\geq v_\infty$, $x_g := v_g - v_\infty \leq \rho_{\text{loc}} x_{g-1} + s_g^2 \to 0$ by the standard perturbed contraction argument (split $\sum_j \rho^{g-j} s_j^2$ at $j = g/2$). If an iterate drops below v_∞: define the unperturbed comparison $u_g = T(u_{g-1})$ started there; then $u_g \uparrow v_\infty$ (monotone, bounded, $T(v) > v$ below v_∞) and $v_g \geq u_g$ by induction ($s_g^2 \geq 0$, T monotone), so $\liminf v_g \geq v_\infty$, while $v_g < v_\infty + s_g^2$ gives $\limsup v_g \leq v_\infty$. Alternating regimes inherit both barriers. Linearizing T at v_∞ gives the local rate.

(ii) Divergence. If $(1-\lambda)(1+c_\infty) > 1$ then, since $e^{2\prime} \geq c_\infty$ everywhere (the concave derivative decreases *to* its limit), $g' \geq (1-\lambda)(1+c_\infty) - 1 > 0$, so g is nondecreasing and $g(v) \geq g(0) > 0$ for all v: $v_g \geq v_0 + g \cdot T(0) \to \infty$, regardless of the schedule ($s_g^2 \geq 0$ only helps).

(iii) Substituting $e^2(w) = e_0^2 + cw$ makes T affine with slope $(1+c)(1-\lambda)$; solve the fixed point. $\square$

B The variance ODE and the orthogonality shield

Lemma 1. One Euler step $x' = \big(1 + (\frac{1}{2} - \hat\kappa/s)\,dt\big)x - (b/s)\,dt + \sqrt{dt}\,\xi$ is affine in x plus independent noise, so $\mathrm{Var}(x') = \big(1 + (\frac{1}{2} - \hat\kappa/s)dt\big)^2 V + dt = V + \big[(1 - 2\hat\kappa/s)V + 1\big]dt + O(dt^2)$; let $dt \to 0$. With $\hat\kappa = \kappa = s/\gamma^2$: $\kappa/s = 1/\gamma^2$ and $V = \gamma^2$ solves the equation exactly (direct check), confirming the bookkeeping. The intercept b moves only the mean. $\square$

Lemma 2. Define $(\hat\kappa, b)$ as the affine $L^2(p_t)$ projection of the estimator itself; the residual $\delta = \hat\varepsilon - \hat\kappa x - b$ satisfies the normal equations $\mathbb{E}_{p_t}[\delta] = \mathbb{E}_{p_t}[x\delta] = 0$ by definition. The first variation of the variance flow under drift perturbation $\eta\,\delta/s$ is $(2/s)\mathbb{E}_{p_t}[x\delta]\,dt = 0$ at $\eta = 0$: no first-order thinning. The leading effect is the accumulated squared displacement $\mathbb{E}[u^2] \geq 0$ of the integrated kicks δ/s, an injection term. (Along the reverse path the ensemble deviates from p_t; the cross term is then $O(\|\hat p_t - p_t\|\,r(t))$, second order in the total perturbation.) $\square$

C Floor lemmas and closed forms

Lemma 3. From (1), $\kappa(t) = s/(a^2\sigma^2 + s^2)$ with $s^2 = 1 - e^{-t}$: $\kappa \to 0$ as $t \to 0$ ($s \to 0$, $\gamma^2 \to \sigma^2$), $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, and maximizing over s gives $\kappa_{\max} = 1/(2\sigma)$ at $s = \sigma$ (i.e. $t \approx \sigma^2$), to leading order in σ^2. Below the band the remaining e-folds are $\int_{t_0}^{t_-} 2\kappa/s\,dt = \int 2\,dt/\gamma^2 = 2\log\frac{\sigma^2 + t_-}{\sigma^2 + t_0} \leq 2\log(1 + \rho^2/4) \leq \rho^2/2$, using $t_- \approx \bar\kappa^2\sigma^4$. $\square$

Lemma 4. Substitute $t = \sigma^2 u$, $V = \sigma^2 W$: $s^2 = \sigma^2 u(1 + O(\sigma^2 u))$, $\gamma^2 = \sigma^2(1+u)(1 + O(\sigma^2 u))$, so $\kappa = (1/\sigma)\frac{\sqrt{u}}{1+u}(1 + O(\sigma^2 u))$ and $2\hat\kappa/s = (2m(u)/\sqrt{u})/\sigma^2$; the ODE becomes $-dW/du = 1 - (2m/\sqrt{u})W + \sigma^2 W(1 + o(1))$. On $u \in [0, U]$ the perturbing terms are $O(\sigma^2 U)$; the band occupies $u \le u_+ = O(1/\rho^2)$, giving the stated $O(\sigma^2/\rho^2)$. For the boundary: the uncapped equation $-dW/du = 1-2W/(1+u)$ has general solution $W = (1+u)+A(1+u)^2$ (direct check), so deviations from the slave solution $W = 1+u$ contract as $\big((1+u)/(1+u')\big)^2$ under downward integration from u' to u; the influence of the $t \gtrsim 1$ boundary layer at band entry is suppressed by this quadratic factor and is absorbed in the same error term. *Numerical check:* the slave solution is invariant to 3×10^{-15} under the integrator; finite-σ floors converge to the limit values from above (e.g. $\rho = 0.39$: 3.953/3.812/3.775 at $\sigma = 0.10/0.05/0.02$ vs. limit 3.769). □

Theorem 2(U). By the scalar comparison principle (equal forcing and data; smaller $\hat\kappa$ means larger V), any profile with $\hat\kappa \le \bar\kappa$ has $V \ge V_{\hat\kappa\equiv\bar\kappa}$, one-sided; no lower bound on $\hat\kappa$ is used. For the constant envelope, (4) with $m \equiv \rho/2$ has integrating factor $e^{-2\rho\sqrt{u}}$: $\frac{d}{du}\big[We^{-2\rho\sqrt{u}}\big] = -e^{-2\rho\sqrt{u}}$, so for data of sub-quadratic growth at $u = \infty$, $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}}du = 2\int_0^\infty xe^{-2\rho x}dx = \frac{1}{2\rho^2}$, exactly, for every $\rho > 0$. $1/(2\rho^2) > 1 \iff \rho < 1/\sqrt{2}$.

Theorem 2(N). The band edges solve $\sqrt{u}/(1+u) = \rho/2$, a quadratic in $x = \sqrt{u}$: $\rho x^2-2x+\rho = 0$, $x_\mp = (1\mp q)/\rho$, $q = \sqrt{1-\rho^2}$ (nonempty iff $\rho < 1$). Above the band the solution is the exact slave $W = 1+u$; it enters the band with $W(u_+) = 1+u_+$. Across the band ($m = \rho/2$), the same integrating factor with antiderivative $F(u) = -(2\rho\sqrt{u}+1)e^{-2\rho\sqrt{u}}/(2\rho^2)$ gives

$$W(u_-) = (1+u_+)e^{-2\rho(x_+-x_-)} + \frac{(2\rho x_- + 1) - (2\rho x_+ + 1)e^{-2\rho(x_+-x_-)}}{2\rho^2},$$

and $2\rho x_\mp = 2(1\mp q)$, $2\rho(x_+-x_-) = 4q$ yield the displayed W_{exit}. Below the band the equation is uncapped with exact general solution $(1+u)+A(1+u)^2$; matching at u_- gives $A = \big(W(u_-)-(1+u_-)\big)/(1+u_-)^2$ and $g(\rho) = W(0^+) = 1+A$, which is (6). Limits: as $\rho\to1^-$ ($q\to0$), $u_\pm\to1$ and $g\to1$; as $\rho\to0$ ($q\to1$), $u_+\approx4/\rho^2$, $u_-\approx\rho^2/4$, and expanding, $g = \big[8e^{-4}+1-5e^{-4}\big]/(2\rho^2)+O(1) = (1+3e^{-4})/(2\rho^2)+O(1)$, giving $C_0 = (1+3e^{-4})/2$. Positivity $g > 1$ on $(0,1)$ follows from $W(u_-) > 1+u_-$ (the band under-contracts relative to the slave). *Numerical check:* (6) and the exact $1/(2\rho^2)$ match direct integration of (4) to $\le 2.4\times10^{-5}$ over $\rho\in[0.1,1]$ (pure Euler residual).

Finally, Lemma 2 extends both bounds from the affine part to the full estimator (the residual only adds variance), and Lemma 4 transfers the limit values to finite σ with the stated error. □

Corollary 1 is Theorem 2(U) read at fixed $\bar\kappa$ with $\sigma\to0$; the capacity bound inverts $1/(2\rho^2) \le C$. □

D Posterior-mean sharpening and per-axis necessity

Proposition 1 (Tweedie sharpening; *we prove*)**.** *In the Gaussian normal coordinate with axis variance σ^2_{ax}, the jump with the exact predictor outputs the posterior mean, with*

$$\operatorname{Var}(\hat x_0) = \frac{a^2\sigma^4_{\text{ax}}}{\gamma^2_{\text{ax}}(\hat t)}, \qquad \frac{\operatorname{Var}(\hat x_0)}{\sigma^2_{\text{ax}}} = \frac{a^2\sigma^2_{\text{ax}}}{a^2\sigma^2_{\text{ax}}+s^2(\hat t)} < 1,$$

and with a trained predictor $\hat\varepsilon = \varepsilon^* - \tilde\delta$, $v_{\text{jump,ax}} = a^2\sigma^4_{\text{ax}}/\gamma^2_{\text{ax}} + (s^2/a^2)\mathbb{E}[\tilde\delta^2] + 2(s/a)\operatorname{Cov} + O(s^4/\tau^2)$.

Proof. $\hat x_0 = (x - s\varepsilon^*)/a = x(1-s^2/\gamma^2)/a = (a\sigma^2_{\text{ax}}/\gamma^2)x$, so $\operatorname{Var} = a^2\sigma^4_{\text{ax}}/\gamma^2$; Tweedie's formula identifies this with $\mathbb{E}[X_0 \,|\, X_{\hat t}]$ [14]. The error term is the jump map applied to $\varepsilon^* - \tilde\delta$; the curvature term is the posterior-mean chord bias. □

Consequences: strongly singular axes ($\sigma_{\text{ax}} \ll s$) are collapsed onto the manifold (the matcher must top up); fat axes ($\sigma_{\text{ax}} \gg s$) are untouched while injection fattens them (the matcher must contract); axes with $\sigma_{\text{ax}} \sim s$ sharpen by an axis-dependent factor strictly inside $(0,1)$. Hence the required correction is signed and anisotropic: any correct matcher must be bidirectional and per-axis. (*We observe:* scalar matching leaves thin ratios at 0.62 to 0.65 on MNIST latents; per-axis matching achieves $|\text{thin} - 1| \leq 0.012$.)

E The residual channel (*we model*)

The orthogonal residual δ has measured band risk $r^2_{\text{res}}(t) \approx 0.02$ to 0.05, flat in t and in sample size, with path-coherence time $\approx \nu t$, $\nu \approx 0.75$. Under a block-correlation model (coherent within blocks of length νt, independent across blocks, no systematic over-contraction), the injected variance obeys $\Phi_0(t_0) \geq c\,\nu\rho_0(\sigma^2 + t_0)$: decreasing in t_0 until $t_0 \sim \sigma^2$, then saturating, the same shape as the measured excess (knee at $t_0 \approx \sigma^2$, n-flat), with the single constant effective to within a factor ~ 2. In this paper the channel plays a supporting role only: it is the measured $\sim 30\%$ single-pass remainder on top of the closed-form floor, which does not depend on it.

F Experimental details

Networks and training. 3-layer MLPs, width 256, SiLU, 32-dim sinusoidal time embedding; Adam, learning rate 2×10^{-3}; batch 512 (ring) / 256 (MNIST); denoising score matching with $t \sim U(10^{-3}, 8)$; fresh network each generation.

Sampling. Reverse Euler-Maruyama on a 200-point geometric grid from $T = 8$; the ablation zeroes the $\sqrt{|dt|}\,\xi$ term only. Jump: $\hat{t} = 0.05$.

Anchor. Ring: radial (normal) bidirectional match of the pool's radial variance to the stale reference's, contract-by-scaling / top-up-by-noise. MNIST: per-point kNN-PCA ($k = 96$, $k_{\text{dim}} = 12$) charts from the reference; normal coordinates rescaled to the reference's local normal eigenvalues in the same frames. Reference: $m = 2{,}000$ real samples drawn once at $g = 0$, never refreshed (provenance-free: the matcher is applied to the whole pool without labels).

Measured network profile. The fitted slope $\hat{\kappa}(t)$ is extracted by affine regression of $\hat{\varepsilon}$ against x on fiber lines at each t; it tracks $\kappa(t)$ (0.95 to 0.97 for $t \geq 0.1$), saturates at $\bar{\kappa} \approx 3.9$ inside the band, and never exceeds κ (n-independent across 1k to 16k). The decomposition identity $(\kappa - \hat{\kappa})^2\gamma^2 + r^2_{\text{res}} = r^2_{\text{tot}}$ checks numerically along the path.

Anatomy of c. Training single-width tubes at $w \in \{0.0009, \ldots, 0.0169\}$ and measuring $dv/dw - 1$ gives net $c \in \{0.00, 0.05, 0.13, 0.10, 0.05\}$ (small, positive, concave) as the sum of a deficit response ≈ -0.68 (a fatter pool needs a smaller peak slope) and an injection response $\approx +0.73$ (a fatter pool damps injected variance less). The same probe's per-width slope fits give the capacity-degradation law $\bar{\kappa}(w) = 4.43 - 11.6\sqrt{w}$ used in Section 5, which refines this anatomy into three channels: writing $\frac{d}{dw}\Phi_{\text{det}}(w, \bar{\kappa}(w)) = \partial_w \Phi_{\text{det}}|_{\bar{\kappa}} + (\partial_{\bar{\kappa}}\Phi_{\text{det}})\,\bar{\kappa}'(w)$, the mid-range values are a deficit ≈ -0.94, a capacity-degradation term $\approx +0.7$, and a residual injection $\approx +0.3$ to $+0.4$; their sum reproduces the measured net c pointwise ($+0.13$ derived vs. $+0.13$ measured at $w = 0.0049$; $+0.10$ vs. $+0.10$ at $w = 0.01$). Most of the apparent injection response is thus the degradation of $\bar{\kappa}$ on fatter pools, and the residual injection is stable and positive, as the residual-channel model (Appendix E) requires. We trust this decomposition mid-range only: at the thinnest width the $\sqrt{w}$ parametrization makes $\bar{\kappa}'(w)$ diverge, an artifact of the fit, not a mechanism.